15. $\frac{20}{\pi^2}$H inductor is connected to a capacitor of capacitance C. The value of C in order to impart maximum power at 50 Hz is

(a) 50 μF (b) 0.5 μF

(c) 500 μF (d) 5 μF

Answers

1) a	**2)** d	**3)** b	**4)** d	**5)** a
6) a	**7)** a	**8)** b	**9)** c	**10)** a
11) c	**12)** c	**13)** d	**14)** c	**15)** d

II Short Answer Questions

1. What is meant by electromagnetic induction?
2. State Faraday's laws of electromagnetic induction.
3. State Lenz's law.
4. State Fleming's right hand rule.
5. How is Eddy current produced? How do they flow in a conductor?
6. Mention the ways of producing induced emf.
7. What for an inductor is used? Give some examples.
8. What do you mean by self-induction?
9. What is meant by mutual induction?
10. Give the principle of AC generator.
11. List out the advantages of stationary armature-rotating field system of AC generator.
12. What are step-up and step-down transformers?
13. Define average value of an alternating current.
14. How will you define RMS value of an alternating current?
15. What are phasors?
16. Define electric resonance.
17. What do you mean by resonant frequency?
18. How will you define Q-factor?
19. What is meant by wattles current?
20. Give any one definition of power factor.
21. What are LC oscillations?

III Long Answer Questions

1. Establish the fact that the relative motion between the coil and the magnet induces an emf in the coil of a closed circuit.
2. Give an illustration of determining direction of induced current by using Lenz's law.
3. Show that Lenz's law is in accordance with the law of conservation of energy.
4. Obtain an expression for motional emf from Lorentz force.
5. Give the uses of Foucault current.
6. Define self-inductance of a coil interms of (i) magnetic flux and (ii) induced emf.
7. How will you define the unit of inductance?
8. What do you understand by self-inductance of a coil? Give its physical significance.
9. Assuming that the length of the solenoid is large when compared to its diameter, find the equation for its inductance.

UNIT-4(XII-Physics_Vol-1).indd 260 17-02-2020 13:20:35